

SmartAgri

Smart Agricultural Support System

CST 394-2

GROUP 07

COMPUTER SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

UVA WELLASSA UNIVERSITY OF SRI LANKA

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Introduction



What is the System ?

A web-based platform designed to help farmers, entrepreneurs, and IT professionals manage agricultural activities smarter and more efficiently.

Why This Project?

- Traditional farming decisions are mostly based on experience
- New entrepreneurs lack agricultural knowledge
- Difficulty in predicting profits and crop performance
- Limited digital connection between farmers and traders
- Lack of an all-in-one digital platform for agribusiness management, prediction, tracking, and trading activities

Problem

⚠️Lack of agricultural knowledge for new entrepreneurs

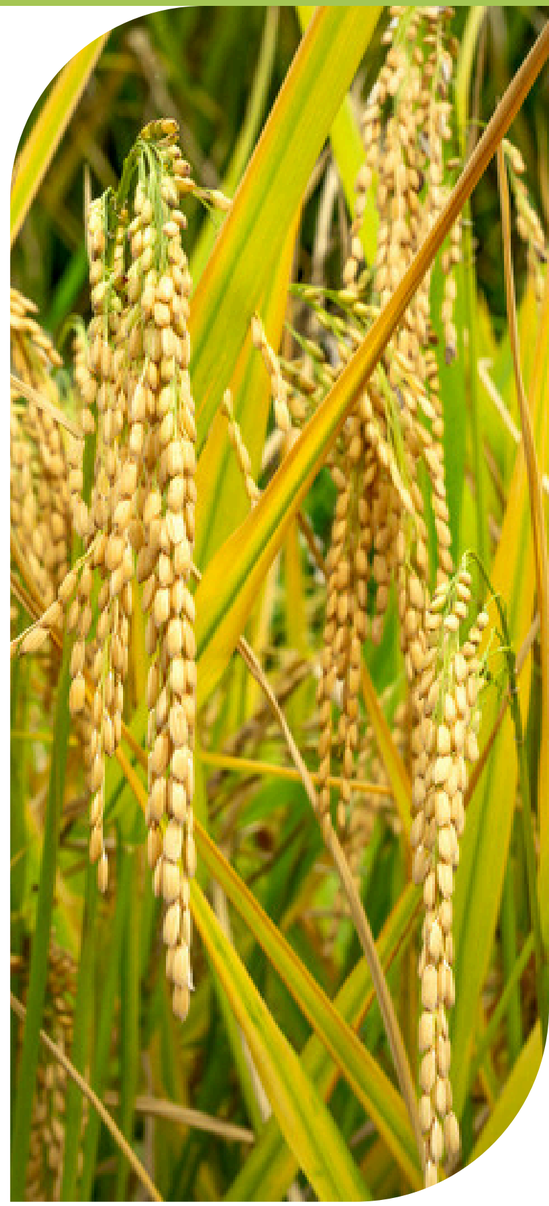
⚠️ Poor crop selection and decision making

⚠️ No proper yield and profit prediction systems

⚠️ Difficulty connecting farmers with buyers

⚠️ Manual and inefficient farm management processes







Solution

✅ Crop Recommendation System
Suggests the best crops based on soil, season, and environment

✅ Yield & Profit Prediction
Uses AI/ML models to estimate productivity and profitability

✅ Digital Marketplace
Connects agribusiness managers directly with traders

✅ Farm Management System
Manage farm activities, crop records, and performance digitally

✅ Role-Based Access
Separate access for Admin, Agribusiness Manager, and Trader

✅ Smart Calendar Tracking
Track planting, watering, fertilizing, and harvesting schedules easily

Novelty Features

✅AI Crop Recommendation
Suggests suitable crops based on farm and environmental data.

✅Yield & Profit Prediction
Predicts expected harvest and estimated profit.

✅Smart Farming Calendar
Helps manage and track farming activities and schedules.

✅Multi-Farm Management
Allows users to manage multiple farms in one system.

✅All-in-One Agribusiness Platform
Combines prediction, management, and trading features into one platform.



Aim

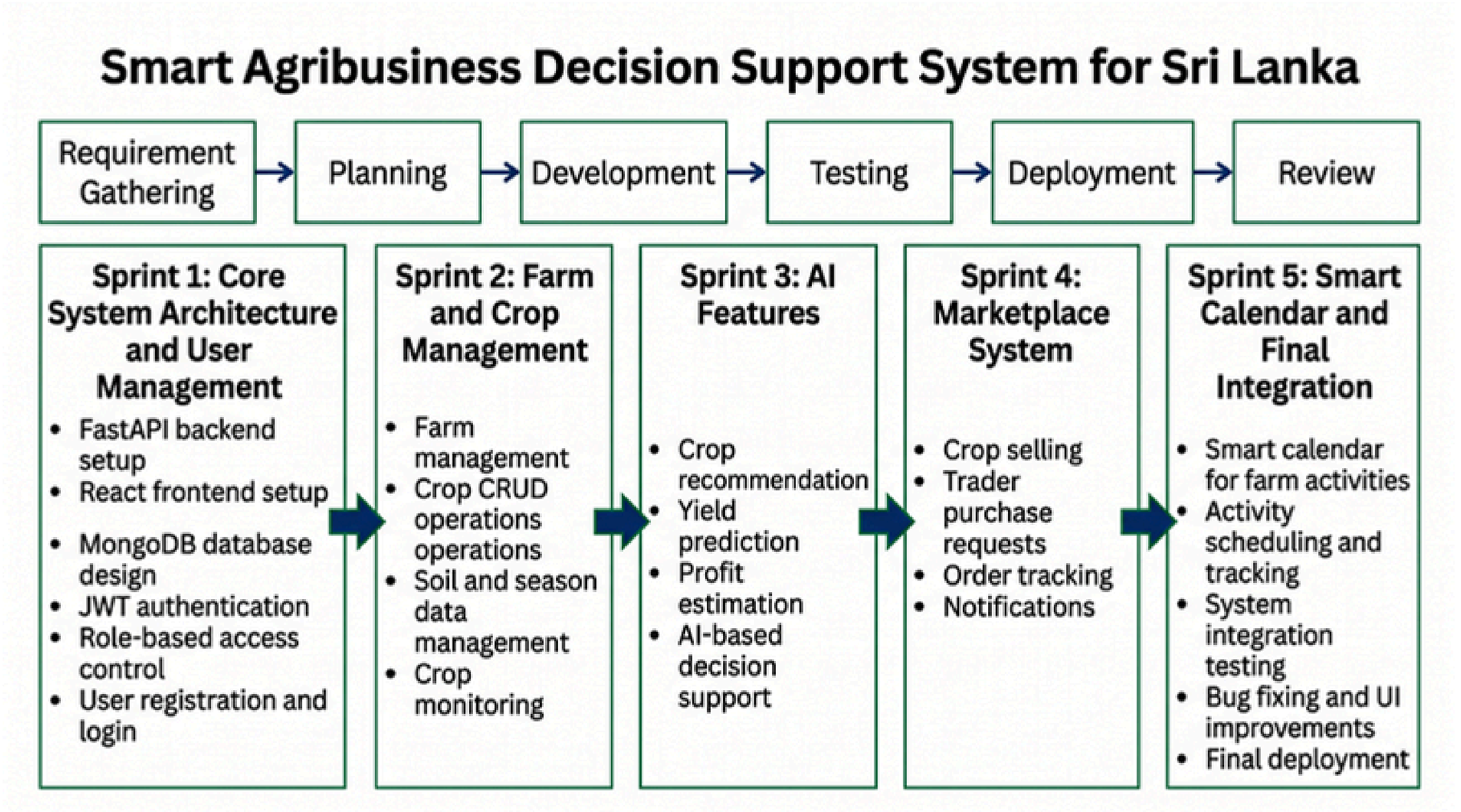
- Smart web platform connecting farmers and buyers while improving agribusiness decisions and productivity in Sri Lanka.



Objectives

- To develop an AI-powered crop recommendation system using soil, season, and location data.
- To predict crop yield and profit using machine learning models.
- To Build a smart farm and crop management system for tracking activities and records.
- To create a digital marketplace connecting agribusiness managers and traders.
- To Implement secure role-based access control for all users.

Methodology (Agile methodology, Scrum framework)



Functional Requirements

Farm Management

Manage farm details, soil data, and farm activities

Crop Management

Add, update, categorize, and track crops

AI Features

Crop recommendations, yield prediction, and profit estimation

Marketplace System

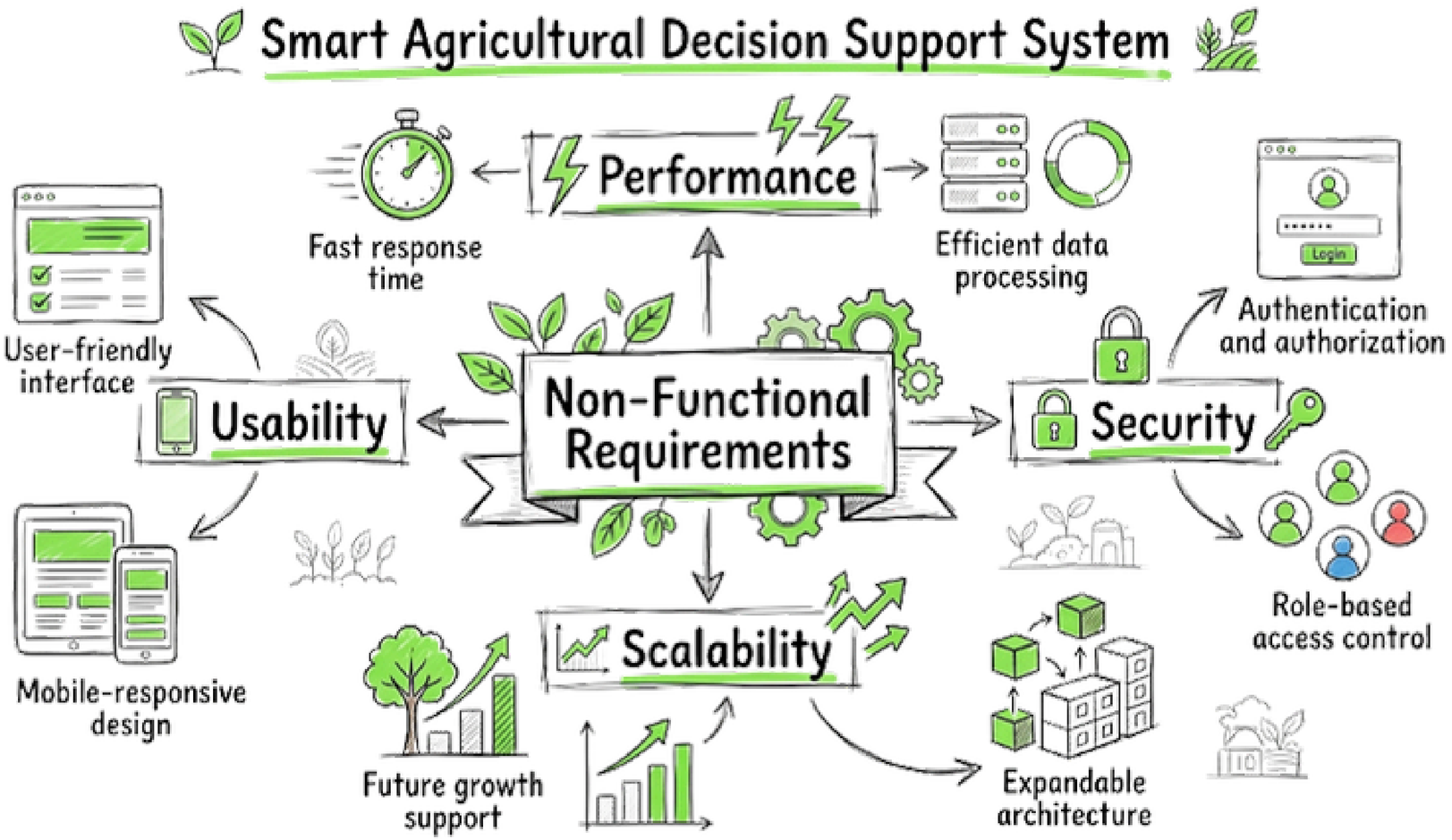
Buy, sell, and manage crop transactions between Agribusiness managers and traders

User Management

User registration, login, role management, and profile handling



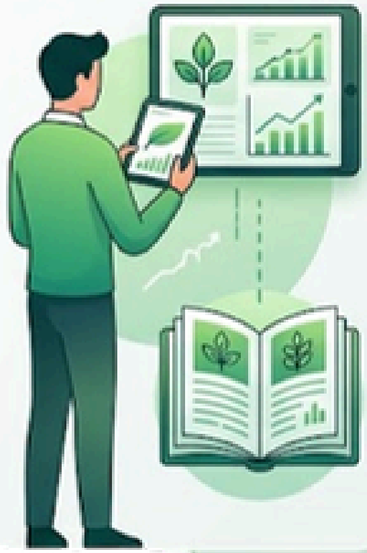
Non- Functional Requirements



User Roles

Visitor

- Access crop recommendations
- View farming knowledge



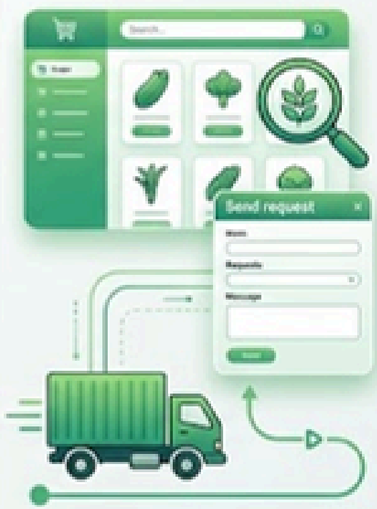
Agribusiness Manager

- Manage farms and crops
- View AI predictions
- Handle sales and orders



Trader

- Search available crops
- Send purchase requests
- Track order status

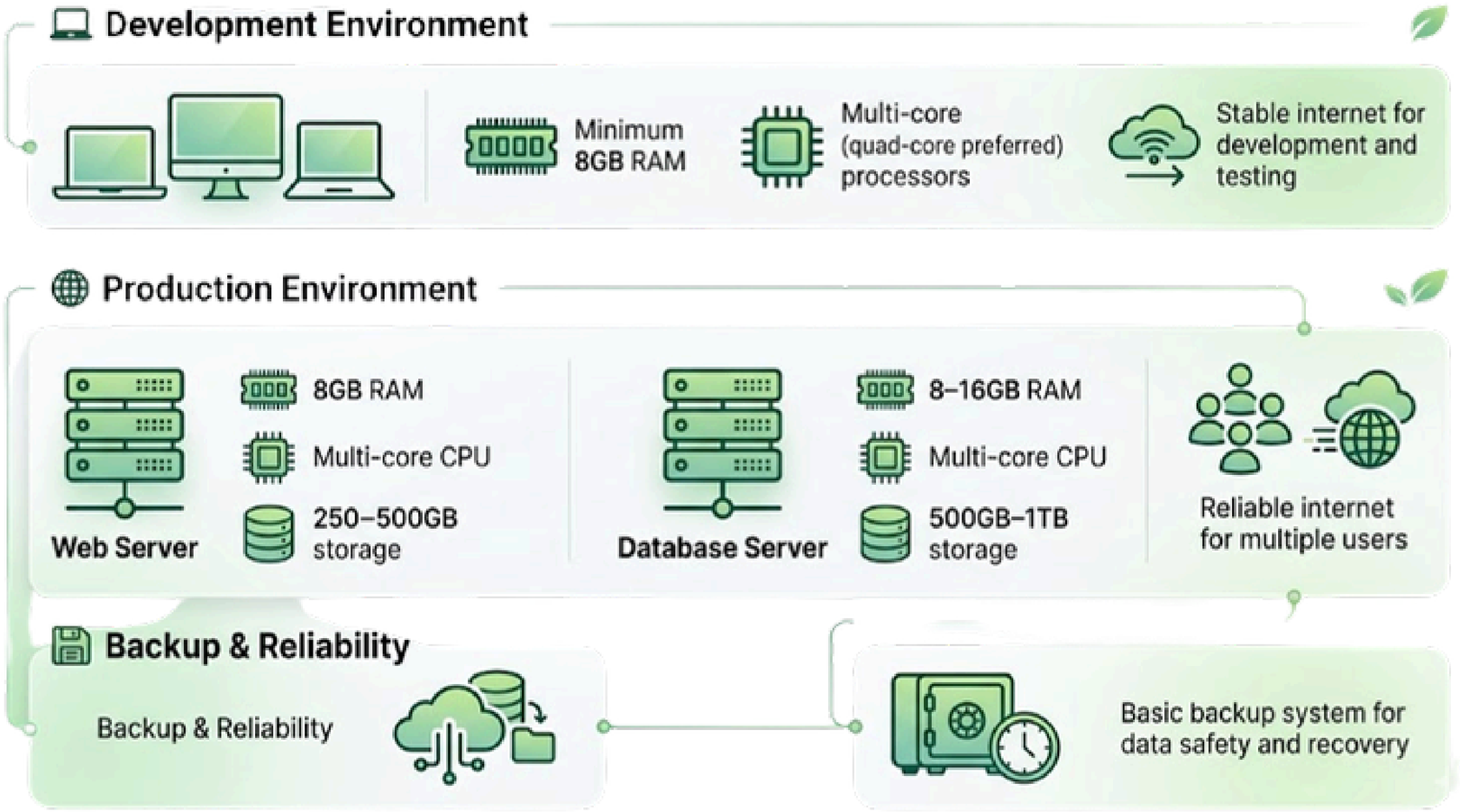


Admin

- Manage users and roles
- Monitor system activities
- Maintain security



Hardware Requirements



Software Requirements

Development Tools



VS Code
IDE



GitHub
Version Control System



Postman
API Testing Tool

Client Environment





Access via modern web browsers





Access via modern web browsers

Technologies

FRONT-END



- React.js
- Tailwind CSS

BACK-END



- FastAPI (Python)
- RESTful API Architecture

DATABASE



- Mysql

DEPLOYMENT



- Cloud Deployment (AWS / Render / Railway / Azure)

Gantt Chart

Task	Week														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Client Interaction & Requirement Refinement															
Initial Planning & System Setup															
UI/UX Design (Iterative)															
Core Development (Modules)															
Continuous Testing & Debugging															
Feature Enhancements (Based on Feedback)															
Integration & System Testing															
Deployment Preparation															
Final Deployment & Maintenance															
Documentation (Continuous)															

Meet Our Team



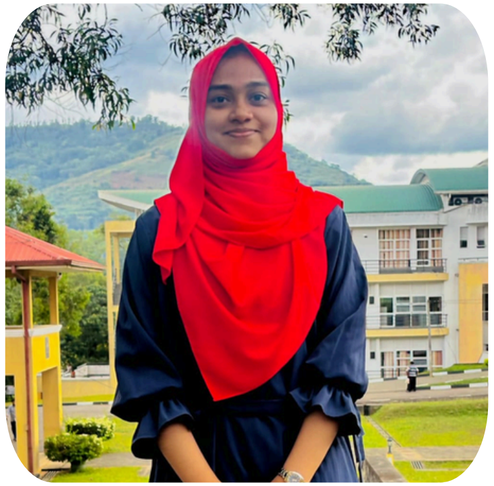
UWU/CST/22/010
M.R.T Gimhan



UWU/CST/22/013
S.D.T Samarawickrama



UWU/CST/22/021
P.I Ihalavithana



UWU/CST/22/068
F. Nuha

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Thank You